

# SREE GNANESH DOMMETI

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## Education

### University of Maryland, College Park

*M.S. in Applied Machine Learning; GPA: 3.88 / 4.0*

May 2026

College Park, MD

### National Institute of Technology, Calicut

*B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10*

May 2022

Calicut, India

## Experience

### Qualcomm

July 2022 – July 2024

*Software Engineer*

*Hyderabad, India*

- Maintained and debugged production LTE Layer 3 RRC modules including Cell Selection, Paging, Security, Handover, and System Information; designed fuzz tests for malformed OTA messages and fake base-station behavior.
- Resolved 100+ customer, field, and regression issues by analyzing modem logs, debugging crashes, and landing targeted fixes with <48h turnaround on high-priority escalations.
- Delivered feature enhancements for CRS/AUX capture, MPS priority indication, OEMPD refactoring, and multivector/EFS logging, improving diagnostics and carrier/OEM compliance.
- Built Python automation tools for changelist assignment, regression tracking, priority assignment, and RRC ownership routing; reduced manual engineering effort by 40%.

### FROOT Lab, University of Maryland

November 2025 – Present

*Research Assistant, Advisor: Alan Zaoring Liu*

*College Park, MD*

- Worked on ProjectASAP, implementing research-backed systems components for workload analysis, sketch planning, streaming summaries, and approximation-aware query serving.
- Built software paths across Prometheus `remote_write`, Arroyo, Kafka, and Rust query services, helping move the project from research prototype toward reproducible telemetry benchmarks.
- Additional NetArena work: built a Dockerized MALT purple agent with LiteLLM prompting, output validation, retry logic, and local benchmark submission workflows.

### GAMMA Lab, University of Maryland

January 2026 – Present

*Research Assistant, Advisor: Dinesh Manocha*

*College Park, MD*

- Building evaluation tooling for **TraceGround**, checking intermediate MLLM reasoning steps against grounded visual/textual evidence.

### Connycat

June 2025 – August 2025

*ML Engineer Intern, CampusAI*

*College Park, MD*

- Built FastAPI services and Elasticsearch retrieval for a campus-event recommendation backend; normalized event feeds and served personalized results at <300 ms p95 retrieval and <600 ms p95 end-to-end latency.
- Containerized services with Docker, deployed on AWS EC2, integrated MLflow experiment tracking, and implemented collaborative-filtering pipelines over implicit interaction data.

## Projects

### eXpOS - Experimental Operating System Kernel | C, XSM Assembly, OS Internals

- Implemented process, memory, file, device, and resource-management modules on the XSM simulator, enabling multitasking, timer-driven context switching, and UNIX-like process execution.
- Built demand paging with page tables and page-fault handling; implemented 19 interrupts/system calls including fork, exec, wait, signal, semaphores, terminal I/O, and disk operations.

### LLM Inference Benchmarking Harness | Python, Docker, AWS EC2 GPU, vLLM, SGLang

- Built a load-testing suite for Mistral-7B-Instruct-AWQ with Poisson arrivals, 1-64 concurrent users, and metrics for latency, throughput, token cost, GPU utilization, and saturation behavior.
- Measured SGLang at 4,001 tokens/sec saturation and vLLM with 34% better throughput at 16 users, producing deployment guidance under randomized production-like traffic.

### ProjectASAP - Streaming Telemetry Query System | Python, Rust, Arroyo, Kafka, Prometheus, Docker

- Implemented workload-analysis and sketch-planning services for high-cardinality telemetry queries, using configurable sketch choices and transparent fallback to Prometheus for unsupported query paths.
- Connected Prometheus `remote_write` ingest, Arroyo streaming jobs, Kafka sketch transport, and Rust query-serving components into an approximation-aware telemetry pipeline.

## Skills

**Languages:** C, C++, Python, SQL, JavaScript, Rust, Dart

**Backend / Systems:** FastAPI, Node.js, Express.js, REST APIs, Linux, OS internals, concurrency, debugging, protocol stacks, fuzz testing

**Data / Infra:** PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Kafka, Prometheus, Grafana, Docker, Kubernetes, AWS EC2/ECS/S3/Lambda, Terraform

**Tools:** Git, GitHub Actions, Jenkins, Perforce, Jira, Postman, MLflow, DVC, OpenTelemetry